JMod Inputs

List of search parameters available to run JMod with. The path to the mzML and the path to the spectral library are required for the search to start running.

Param	Description	Type	Default	Example
-i, --mzml	Path to mzML file to be searched	str		--mzml "/path/to/file.mzML"
-l, --speclib	Spectrum library to search (library format)	str		--speclib "/path/to/lib.tsv"
--config_json	JSON configuration parameter file to run the search with (./data/default_config.json)	str	None	--config_json "/path/to/config.json"
-o, --output_folder	Output folder for search results. Defaults to the directory of the raw files.	str	None	--output_folder "/path/to/folder"
--ms1_ppm	MS1 matching tolerance in ppm (If 0, optimized value will be calculated).	int	0	--ms1_ppm 10
-p, --ppm	MS2 matching tolerance in ppm.	int	10	--ppm 15
--rt_tol	User defined retention time tolerance (If 0, optimized value will be calculated.)	int	0	--rt_tol 20
--user_rt_tol	Use user-defined retention time tolerance	bool	False	--user_rt_tol
-m, --atleast_m	Required number of fragments matched from 10 most intense library fragments	int	3	--atleast_m 5
--lib_frac	Fraction of library intensity required to be matched	float	0.5	--lib_frac 0.3
--iso	Use MS2 isotopes in search	bool	False	--iso
--num_iso	Number of MS2 isotopes to consider (if MS2 isotopes being used)	int	2	--num_iso 3
--no_ms1_req	Do not require the observation of an MS1 peak for consideration in the search.	bool	True	--no_ms1_req
--tag	Tag used in experiment, if any. See jmod/src/MassTags for list of available tags	str	None	--tag "PSMtag_5plex"
--plexDIA	Use plexDIA mode for search	bool	False	--plexDIA
--timeplex	Use timePlex mode for search	bool	False	--timeplex
--num_timeplex	Number of time offset injections for timePlex search	int	0	--num_timeplex 2

Library Format

Definition of columns required for tab separated JMod library. Each row represents a fragment. Rows are grouped by columns *ModifiedPeptide* and *PrecursorCharge* into precursors.

Column	**Definition**	**Req/Opt**
ModifiedPeptide	Peptide sequence including modifications	Req
StrippedPeptide	Peptide sequence not including modifications	Req
PrecursorCharge	Charge of the precursor	Req
RT	Retention time of the precursor	Req
PrecursorMz	m/z of the precursor	Req
FragmentMz	m/z of the fragment	Req
RelativeIntensity	Intensity of the fragment	Req
FragmentType	Fragment type (b, y)	Req
FragmentCharge	Fragment charge (1, 2, etc.)	Req
FragmentSeriesNumber	Fragment series number (12, 9, 5, etc.)	Req
FragmentLossType	Fragment loss type (noloss, NH3, H2O)	Opt
ProteinID	All proteins that contain the fragment’s precursor	Opt
ProteinGroup	Inferred proteins	Req
ProteinName	Names of proteins in ProteinGroup	Opt
Genes	Gene names mapping to proteins in ProteinGroup	Opt
IonMobility	Ion mobility of the precursor	Opt

JMod Test Mode Parameters

Param	Description	Type	Default	Example
--test_mode	Use test mode to run a dry run of search	bool	False	--test_mode
--test_rt_min	Minimum RT threshold used in test mode	float	0	--test_rt_min 20
--test_rt_max	Maximum RT threshold used in test mode	float	10	--test_rt_max 40
--test_mz_min	Minimum m/z threshold used in test mode	float	400	--test_mz_min 200
--test_mz_max	Maximum m/z threshold used in test mode	float	800	--test_mz_max 600